The Exemplar Explained
Cybersecurity Incident Report: Network Traffic Analysis

Part 1: Provide a summary of the problem found in the DNS and ICMP traffic log	Explanation
A. As part of the DNS protocol, the UDP protocol was used to contact the DNS server to retrieve the IP address for the domain name of yummyrecipesforme.com. The ICMP protocol was used to respond with an error message, indicating issues contacting the DNS server. B. The UDP message going from your browser to the DNS server is shown in the first two lines of every log event. The ICMP error response from the DNS server to your browser is displayed in the third and fourth lines of every log event with the error message, “udp port 53 unreachable.” Since port 53 is associated with DNS protocol traffic, we know this is an issue with the DNS server. Issues with performing the DNS	A. From my analysis, I observed that the UDP protocol was used as part of the DNS process to contact the DNS server and retrieve the IP address for the domain name yummyrecipesforme.com. However, instead of receiving a normal DNS response, the ICMP protocol was used to return an error message. This indicates that there was a problem communicating with the DNS server. B. Looking closely at the tcpdump logs, the first two lines of each log event show the UDP packets being sent from the client (browser) to the DNS server. The third and fourth lines show the ICMP error responses sent back to the client. These responses include the message “UDP port 53 unreachable.” Since port 53 is specifically used for DNS services, this clearly points to a problem with the DNS server. Additionally, the plus sign (+) after the query ID (35084) and the “A?” symbol indicate that the system was attempting standard DNS query operations, but they were not successful. C. Based on the repeated ICMP error messages indicating that port 53 is unreachable, it is highly likely that the DNS server is not responding to

protocol are further evident because the plus sign after the query identification number 35084 indicates flags with the UDP message and the "A?" symbol indicates flags with performing DNS protocol operations. C. Due to the ICMP error response message about port 53, it is highly likely that the DNS server is not responding. This assumption is further supported by the flags associated with the outgoing UDP message and domain name retrieval.	requests. This could be due to the DNS service being down, misconfigured, or blocked. The presence of DNS query flags alongside failed responses further supports the conclusion that the domain name resolution process was unsuccessful.

Part 2: Explain your analysis of the data and provide at least one cause of the incident	Explanation
D. The incident occurred today at 1:24 p.m. E. Customers notified the organization that they received the message	D. The incident was identified to have occurred at approximately 1:24 p.m., based on the timestamps observed in the traffic logs. This helps establish a clear timeline for when the issue began affecting users.

"destination port unreachable" when they attempted to visit the website yummyrecipesforme.com. F. The cybersecurity team providing IT services to their client organization are currently investigating the issue so customers can access the website again. G. In our investigation into the issue, we conducted packet sniffing tests using tcpdump. In the resulting log file, we found that DNS port 53 was unreachable. H. The next step is to identify whether the DNS server is down or traffic to port 53 is blocked by the firewall. I. DNS server might be down due to a successful Denial of Service attack or a misconfiguration.	E. Customers reported receiving the error message "destination port unreachable" when attempting to access the website yummyrecipesforme.com. This indicates that their requests were not successfully reaching the required service on the server. F. In response to these reports, the cybersecurity team responsible for IT services began investigating the issue to restore normal access to the website as quickly as possible. G. As part of the investigation, packet sniffing was conducted using tcpdump to analyze network traffic. The results showed that DNS requests sent over UDP port 53 were failing, and the logs consistently returned ICMP error messages indicating that port 53 was unreachable. This confirmed a breakdown in DNS communication. H. The next step in the investigation is to determine the root cause of the issue. Specifically, this involves verifying whether the DNS server is down or if network controls, such as a firewall, are blocking traffic to port 53. I. One possible cause of the incident is that the DNS server is unavailable due to a successful Denial of Service (DoS) attack, which could overwhelm the server and prevent it from responding to requests. Another possible cause is a misconfiguration of the DNS service or firewall settings, which could block or prevent access to port 53.